Uses of Artificial Intelligence in Higher Education

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Abstract: This paper investigates the distribution and focal areas of research being conducted in higher education between the years of 2019 and January, 2023. The analysis examined 723 articles resulting from a structured search of the Gale Onefile, ERIC, and Education Full Text databases. The search strategy included peer-reviewed publications published after 2019, filtered for key terms that included artificial intelligence, higher education, and teaching. A total of 113 articles met inclusion criteria and were coded as belonging to one of the following categories: adaptive systems, assessment and evaluation, profiling and prediction, or intelligent tutoring. The results appear to demonstrate rapid development, but few tools are readily available or usable in classroom contexts at this point in time, more applications are being used for profiling and predict student performance.

Introduction

The field of artificial intelligence (AI) has recently witnessed remarkable advancements, drawing attention from policymakers, artists, educators, and expert commentators. The remarkable human-like proficiency of generative AI agents, across diverse domains have begun to transform modes of learning and working with impressive speed and results. The level of performance large language models (LLMs) alone has improved dramatically over the past 2 years, to the point where GPT4 agents have demonstrated genius level test results in computer science, law, creative writing, and medicine (Kung et al., 2023). Examples of LLMs that are available to the public include variants of Generative Pretrained Transformers (or GPTs) Each of the variants are capable of producing computer code, entire essays, spreadsheets, and data visualizations in response to a text 'prompt' entered by a person. Artificial intelligence is also used to make it possible for computers to 'see' and understand the environment visually. When generative AI is combined with computer vision, remarkable original art, that imitates artistic styles of traditional, cartoon, or anime artists. Some tools available online have not only generated photo-realistic images of actual individuals, but even imitations of their voices speaking and singing. The emergence of these tools has caused controversy, lawsuits, legislative interventions (Chae, 2020) and considerable uncertainty regarding their appropriate use (Cuéllar & Huk, 2022).

The use of AI agents in education is growing at a more cautious rate than the broader AI landscape, educators are grappling with complications arising from the use of these tools in the classroom. Many educators are undecided

whether to disavow generative AI agents as another form of plagiarism, or to employ their use in teaching activities as intelligent assistants. The potential of AI in primary and secondary school education (K-12 in the United States) has been noted in some novel research in pedagogy, administration, and academic integrity, but the greatest impact may come from leveraging AI as a teaching resource (Alam, 2021). While there appears to be considerable uncertainty surrounding AI in education, the benefits may include adaptive learning solutions, automated grading and feedback, and earlier identification of at-risk students. Research in this area demonstrates a number of AI agents that function as either an interactive application or intelligent tutor in particular (Zawacki-Richter et al., 2019).

The breadth of opportunity is accompanied by significant ethical challenges. While plagiarism is a more immediate ethical challenge, there may be equally substantial risk to the students themselves, because AI agents rely on extensive data collection and curation of student information. In higher education, the implications of AI-driven pedagogical advancements may diminish the demand for teaching assistantships and other university positions that serve as vital sources of student financial support. The potential consequences AI systems with superior proficiency and potential has generated considerable uncertainty in terms of future employment opportunities for college-educated individuals, and the nature of work itself.

The current research is an independent investigation that extends a study conducted by Zawacki-Richter et al. in 2019. The goal of the 2019 research was to investigate research trends that have emerged as artificial intelligence, and the applications that are employed in higher education by conducting a systematic review to define the uses of AI in higher education between 2007 and 2018. The study identified 146 articles that met specific inclusion criteria and highlighted the need for further exploration of AI, its applications, and ethical implications in higher education. The objective of the present study is to define the uses of AI in higher education since 2018. To that end, a systematic review of peer-reviewed research was conducted to investigate the scope of AI applications in higher education, identify the journals and geographical locations of the research, and detect any noticeable shifts in research trends in the last four years.

The Study

The present study identified 113 articles acquired from three databases: ERIC (Proquest), Educator's Reference Complete (Gale OneFile), and Education Full Text (EBSCO). These databases were selected to capture works from a diverse spectrum of educational publications, including Resources in Education (RIE) and the Current Index to Journals in Education (CIJE).

The Search String

The researchers modeled a search string that focused on higher education, search terms corresponding to lower grades were not included. Specifically, the terms "K-12", "kindergarten*", "primary school*", "middle school*", "high school*", and "elementary school*" not included, resulting in the search string in Table 1.

Artificial Intelligence AND	artificial intelligence OR machine intelligence OR intelligent support OR intelligent virtual reality OR chat bot* OR machine learning OR automated tutor OR personal tutor* OR intelligent agent* OR expert system OR neural network OR natural language processing
Education level AND	higher education OR college* OR undergrad* OR graduate OR postgrad* OR corporate training* OR professional training* OR vocational education OR adult education
Student	learn* OR student*

Table 1. search string strategy specific to higher education.

Inclusion and exclusion criteria

The search was constrained to include only those publications that have been peer-reviewed and published between January 2019 and January 2023. The search criteria were designed to narrow the set of articles to be reviewed to those limited to some form of artificial intelligence (e.g. machine learning, supervised, or unsupervised

deep learning) used in higher education (see Table 2).

Inclusion Criteria	Exclusion Criteria
● Research conducted on topic, process, or system in higher education ● Empirical, or observational primary research ● Research reports the effects of AI on instruction or learning processes ● Publication must be peer-reviewed ● Research must be reported in English ● Research must be published between 2019 and 2023 ● Full-text is available	● Publications that are concerned with K-12 educational settings ● Not primary research, e.g. research using self-reported data, such as collections, interviews or surveys ● Research from sources such as magazines, news articles, and posters ● Publications in languages other than English ● Research published before 2019 ● Abstracts or summaries ● No use of artificial intelligence in education

Table 2. Final inclusion and exclusion criteria.

Process

A team of four researchers (A, B, C, and D) were each paired with one database to screen, with the exception of Educator's Reference Complete, which A and C screened. Each screening effort used the same search string terms (Table 1), limited to peer-reviewed articles published after 2019, and were maintained separately by A, B, C and D. The initial search collected and recorded reference details, abstracts, conclusions, and codes if the descriptions were within the boundaries set by the inclusion and exclusion criteria described in Table 2. Of the articles in the initial dataset, 105 were obtained through Education Full Text, 21 from Educator's Reference Complete, and 73 from ERIC. After removing 16 duplicates, the team reduced the dataset to 183 articles.

The research team also coded the articles that they included from the respective databases. The codes used in this study are borrowed from the work of Zawacki-Richter et al. (2019) and described in detail in the next section. Researcher A reviewed the entire initial dataset and flagged articles that either did not appear to meet inclusion criteria or have an appropriate code. Researcher B reviewed the flagged articles and reached a consensus with A regarding the coding and inclusion criteria of each article. A fifth researcher not involved in the article collection process also reviewed the flagged articles with B and assisted in creating the consensus with A and B. Those articles not flagged by A were reviewed by B and the other coder. Following the review process, 113 articles remained for analysis.

Findings

The number of peer-reviewed articles focusing on AI in education has increased broadly during recent years (The White House, 2022). Figure 1 Shows the distribution of articles across the range of years considered for the current research. The distribution of research appears to be equally divided between eastern and western research institutions, with the United States, China, and Taiwan accounting for 55 of the articles analyzed.

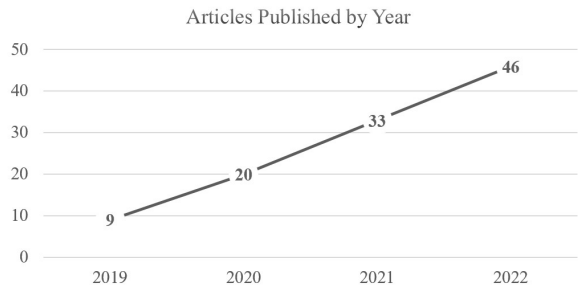


Figure 1. Number of peer-reviewed publications per year, as represented by a sample collected from three educational research resources.

The SJR (SCImago Journal Rank), first introduced in 2008, is a widely recognized open-source ranking system, and based in Scopus as opposed to the Web of Science data that accounts for a journal's 'prestige' (Falagas, 2008). Table 3 lists journals that included the greatest number of publications (n = 79) and their associated rankings, where top-quality journals are scored by their citation potential and grouped by quartile 1 (Q1) through 4 (Q4). The quality of the publications is noteworthy, with 50 of the 79 scoring in the top quartile, 12 in the second quartile, 8 in the fourth, and 9 publications without a SJR score.

Quartile	SJR Rank	Journal	*n*
Q1	1.448	Educational Technology & Society	14
Q1	0.919	Education and Information Technologies	14
Q2	0.454	International Journal of Emerging Technologies in Learning	8
NA	Not Ranked	International Journal of Higher Education Research	5
Q4	0.178	International Educational Data Mining Society	5
		Articles from 43 other journals	*67*

Table 3. Top 3 Journals, with SJR Rank and quartile.

Research categories

Adaptive systems and personalization include AI agents participating in personalization of course content, 'recommender' systems, and knowledge representation. This category was identified as the second most active research area, comprising approximately 20% of the articles (n = 24). It is also noteworthy that research in this category appears to focus on administration (registration, lesson planning, and scheduling) and teaching (tutor, coach or practice). Interestingly, these two areas of research appeared to be grouped regionally, with North American and Western European nations producing more research concerned with administration (Shardlow et al., 2022), whereas research conducted in Asia focused more on teaching (Liu & Ardakani, 2022).

Assessment and evaluation is the third most active research area, comprising approximately 20% of the articles (n = 23). Two major areas of research emerged in this category: (1) automated grading and feedback, and (2) evaluation of student understanding, engagement, and academic integrity. Automated grading research has sought to optimize structured data evaluations (Jescovitch et al., 2021), as well as unstructured data as noted in Automated Essay Scoring (AES) performance (Beseiso et al., 2021). Real-time feedback has been tested by analyzing the explanations provided by students while reading to provide formative feedback using multi-dimensional analysis (Allen et al., 2019).

The *Profiling and prediction* category includes improving student retention decisions, defining academic pathways (course selection and scheduling), mitigating drop-out (retention prediction), student models and academic achievement. This emerged as the largest category of research, comprising approximately 43% of the articles (n = 50). This includes areas such as retention and curriculum personalization, or other methods that seek to enhance student achievement via predictions of learning status, performance, or satisfaction (Ouyang, 2022). In this category, AI agents are employed to make predictions based on student performance measures. The majority of articles were concerned with: (i) predicting student's academic performance (ii) predicting student dropout or retention rate, and (iii) admissions decision-making and course scheduling. Five studies focused on data mining to predict individual performance in introductory undergraduate courses, such as computer science and physics, to identify and reach at-risk students early (Yağcı, 2022). For example, data has been collected from first-year university students and used to predict re-enrollment and graduation rates (Aulck et al., 2019). Other research has investigated the use of admissions records, financial, and academic aid records to predict student dropout rates (Jang et al., 2022), as well as determining educational pathways for student success (T.-C. Yang et al., 2021).

Intelligent tutoring systems are considered participants in teaching by providing suggestions and automated feedback, and is oftentimes referred to as 'blended learning' (Obari, et al., 2020). Intelligent Tutoring Systems (ITS) are subdivided into four categories: (1) teaching course content, (2) diagnosis and feedback of student strengths and gaps, (3) curating learning materials, and (4) facilitating collaboration between learners. ITS was used in teaching English as a foreign language (n = 4), computer science (n = 3), and healthcare (n = 2). The most common form of AI agents is the text-based chat bot that offers personalized recommendations of learning materials to students,

digital study planning, and tools to help students achieve their academic goals (Weber, F. et al., 2022). It is noteworthy that in this context, conversational text-based chat bots are being studied as intelligent tutors that provide supplementary practice for students studying English as a foreign language (Ibna Seraj, & Oteir, 2022). Rather than creating a specialized chat bot intended to teach course content directly, students simply engaged in regular conversation with them in English. For example, Obari, et al. (2020) used Amazon Alexa to provide Japanese speakers practice with speaking in English. All of these studies reported success in improving students' English language skill.

Another interesting example of intelligent tutoring includes research that focuses on analyzing the explanations provided by students while reading. This is an unstructured approach using *Natural Language Processing* (NLP), a number of chatbots have started to appear in research articles, and as productivity tools for programming, content generation, and other business uses. While there is no specific data to support the idea, the researchers believe that the number of articles uncovered in the current research may not be in proportion to their actual use, due to the speed of their adoption (Malik, et., 2021).

Discussion

While artificial intelligence has been noted to be technically difficult to operate, recent developments have made such agents more accessible and better suited for inclusion into classroom activities. While ethical considerations have perhaps made the adoption of AI agents more cumbersome, the underlying software is open-source, which has led to competitive development of chatbots, which has led to an increase in research publications and technical capabilities of the AI agents.

Several questions arise as a result of the study findings. The current research appears to show that there has been an increase in research in the past 5 years. It is interesting to note that there are not many instances of AI agents being used to teach, with the exception of English as a foreign language. It is interesting to note that LLMs are being used to teach language but have also demonstrated a remarkable capability to generate computer code, summarize topics and assess difficult problems in science, and medicine. This raises the question of why there is more activity outside of language education, given the wide variety of potential uses (Hocky & White, 2022).

Another question is the issue of ethics in conducting AI-related research. Remarkably few studies considered the ethical implications of their findings, despite the abundance of ethical considerations as discussed earlier. Overall, it is clear that more research is needed, both in developing the technology to its fullest potential and in considering the ethical impacts that these advancements will have on the students they affect.

Limitations and implications

Although the current research attempts to be as comprehensive and thorough as possible, there are several limitations to the findings presented here. Foremost is that, given the rapidly changing nature of the field, the true scope of the uses of AI in higher intelligence will likely have evolved in a short time following the publication of this work. Additionally, not all efforts to adopt and develop new technologies are recorded as research, so the uses and practices of applications being used to enhance 'productivity' are unknown. The scope of research has increased so much that researchers had to exclude research concerned with primary and secondary school-level practices, which may have inadvertently excluded important developments not within the scope of this review.

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